MILITARY SPECIFICATION SHEET

# CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/174-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

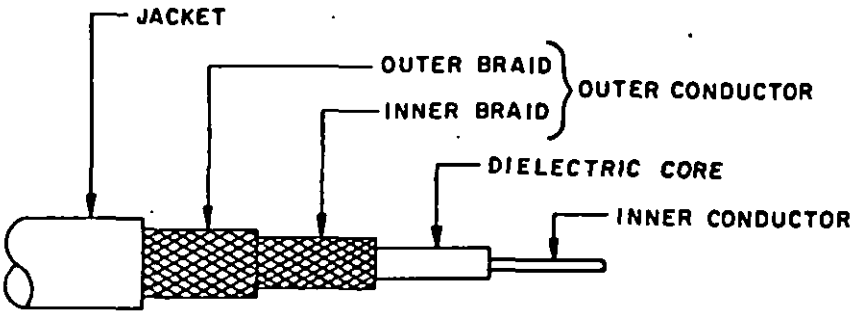



FIGURE 1. Configuration.

TABLE I. Description.

| Components | Construction details |
|---|---|
| Inner conductor | Seven strands of silver-coated copper wire; each strand 0.0312 inch diameter. Overall diameter: .094 inch ±.001. |
| Dielectric core | Type F-1: Solid extruded PTFE. Diameter: .285 inch ±.005. |
| Outer conductor | Double braid of AWG #34 silver-coated copper wire. Diameter: .360 inch maximum. |
| Inner braid | Coverage : 95.7% nominal<br>Carriers : 24<br>Ends : 6<br>Picks/inch: 16.6 ±10% |
| Outer braid | Coverage : 98.0% nominal<br>Carriers : 24<br>Ends : 7<br>Picks/inch: 15.4 ±10% |
| Jacket | Type IX: FEP. Diameter: .390 inch ±.010. |

CAUTION IS DIRECTED TO THE APPLICATION OF THIS CABLE ABOVE 400 MHz. ATTENUATION IS TESTED ONLY AT 400 MHz. SRL AND POWER HANDLING CAPABILITIES ARE NOT STIPULATED HEREIN.

A *denotes changes*



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ENGINEERING INFORMATION:

Continuous working voltage: 1,875 V rms, maximum.

Velocity of propagation: 69.5 percent, *nominal*.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.152 ohm per 100 feet.

Elongation: 20 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in general purpose, high temperature *applications*. (See *connector series* "N" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

*Out-of-roundness*: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 10 pounds, minimum; 50 pounds, maximum.

Aging stability: Not applicable.

*Stress crack resistance*: +230°C ±5°C; mandrel size 7 1/2 times the jacket diameter.

Outer conductor integrity: Not applicable.

Ⓐ Cold bend: -40°C ±2°C.

Dimensional stability: 200°C ±5°C.

Inner conductor from core: 0.250 inch, maximum.

Inner conductor from jacket: 0.312 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Applicable.

Ⓐ Weight: 17.5 pounds/100 feet, maximum.

Electrical:

Continuity: Applicable.

Spark test: 2,000 V rms, +25 percent, -0 percent.

Voltage withstanding: 7,500 V rms, minimum.



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Insulation resistance: Not applicable.

Corona extinction voltage: 2,500 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: 5 dB/100 ft maximum at 400 MHz.

Structural return loss: Not applicable.

Capacitance: 32 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance:. Not applicable.

Phase stability: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

| Part number | Superseded part number |
|---|---|
| M17/174-00001 | RG-393 per MIL-C-17 |

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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